

Rigorous Generalization Study: Transformer-Accelerated VQE Warm-Starting Across Multi-Molecule Families

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Status: Published Research Preprint — DOI: 10.5281/zenodo.22013110

Zenodo Record: zenodo.org/records/22013110

Repository: github.com/aashiq-parinda/quantum-genai-warmstart

Abstract

Variational Quantum Eigensolvers (VQE) require hundreds to thousands of quantum circuit evaluations to converge from random parameter initialization, severely limiting practical application on near-term NISQ hardware. We investigate whether a 13,156-parameter pure-NumPy Transformer encoder trained on Hamiltonian-to-optimal-parameter pairs can generate warm-start initializations that improve VQE convergence, avoid barren plateaus, and generalize across molecular families (H_2 , LiH , BeH_2 , H_4 chain) spanning 4, 6, and 8 qubit systems.

Evaluating across 10 random seeds per experiment with paired t -tests ($\alpha = 0.05$), we find:

- 1. In-Distribution & Interpolation:** Transformer warm-starting achieves a statistically significant $37.8 \pm 4.2\%$ reduction in convergence iterations ($p = 3.4 \times 10^{-5}$) when interpolating along potential energy surfaces of trained molecules (H_2 , LiH).
- 2. Comparison with Classical Hartree-Fock Baseline:** Classical Hartree-Fock (HF) initialization (θ_{HF}) captures $>75\%$ of iteration savings with zero ML training cost. On held-out zero-shot out-of-distribution (OOD) molecules (BeH_2 , H_4 chain), **Hartree-Fock outperforms the Transformer**, achieving $4.1, mHa$ lower ground-state energy and faster convergence ($p = 0.018$).
- 3. Barren Plateau Diagnostic:** Direct gradient variance measurement $Var[\partial E / \partial \theta]$ reveals that while warm-starting maintains large gradient variance on trained systems ($4.1\times$ higher than random), its gradient variance decays sharply towards the barren plateau on 8-qubit H_4 chain ($Var = 0.0049$), whereas Hartree-Fock maintains a robust non-vanishing gradient variance ($Var = 0.0385$, $13.7\times$ higher than random).

We discuss the implications of these empirical null results for quantum machine learning generalization and outline clear limitations.

1. Introduction & Research Problem

1.1 The Bottlenecks of NISQ VQE

Variational Quantum Algorithms (VQAs) optimize a parameterized quantum circuit $|\psi(\theta)\rangle$ to minimize $E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$. Two major barriers hinder practical quantum utility:

1. **Barren Plateau Phenomenon** (McClean et al. 2018): For random initializations $\theta \sim \text{Uniform}[0, 2\pi]^M$, gradient variances decay exponentially $\text{Var}[\partial E / \partial \theta_k] \sim O(2^{-N})$, rendering gradient descent ineffective.
2. **High Sample Complexity**: Evaluating parameter gradients via the Parameter-Shift Rule requires $2M$ circuit measurements per optimization step.

1.2 Generative Parameter Warm-Starting

Generative warm-starting trains a neural network $\mathcal{J}_W(H)$ to map molecular Hamiltonian Pauli representations to near-optimal initialization vectors $\theta_0 = \mathcal{J}_W(H_{\text{new}})$. If successful, warm-starting should start in a non-vanishing gradient basin, reduce convergence iterations, and lower sample complexity.

2. Hypothesis & Falsifiable Diagnostic Criteria

Hypothesis: A transformer encoder $\mathcal{J}_W(H)$ trained on molecular Hamiltonians can generate parameter initializations θ_0 that (1) reduce VQE convergence iterations by $\geq 40\%$ vs random init, (2) outperform classical Hartree-Fock initializations, and (3) maintain non-vanishing gradient variance $\text{Var}[\partial E / \partial \theta]$ across scaling qubit counts N in $\{4, 6, 8\}$.

3. Architecture & Methods

3.1 Unified Hamiltonian Tokenization ($N_{\text{max}} = 8$)

Each molecular Hamiltonian $H = \sum_k h_k P_k$ is tokenized into sequence tokens:

$$\text{token}_k = [\text{one-hot}(\sigma_1^{(k)}), \dots, \text{one-hot}(\sigma_{N_{\text{max}}}^{(k)}), \bar{h}_k]$$

where $\bar{h}_k = h_k / \max_j |h_j|$ normalizes term coefficients. Input dimension $d_{\text{token}} = 4 \times 8 + 1 = 33$, sequence length padded to `max_terms` = 64.

3.2 Transformer Architecture

A pure NumPy 15,436-parameter Transformer Encoder implementing:

- Token Projection: $\mathbb{R}^{33} \rightarrow \mathbb{R}^{32}$

- Multi-Head Attention: $h=2$ heads, $d_k = 16$
- Feed-Forward Block: $32 \rightarrow 128 \rightarrow 32$, ReLU
- Layer Normalization & Global Average Pooling $\mathbb{R}^{32}$
- Output Projection: Dual heads for qubit-pair mask $\mathbf{m}$ in $[0, 1]^{28}$ and parameters $\boldsymbol{\theta}_0$ in $\mathbb{R}^{16}$

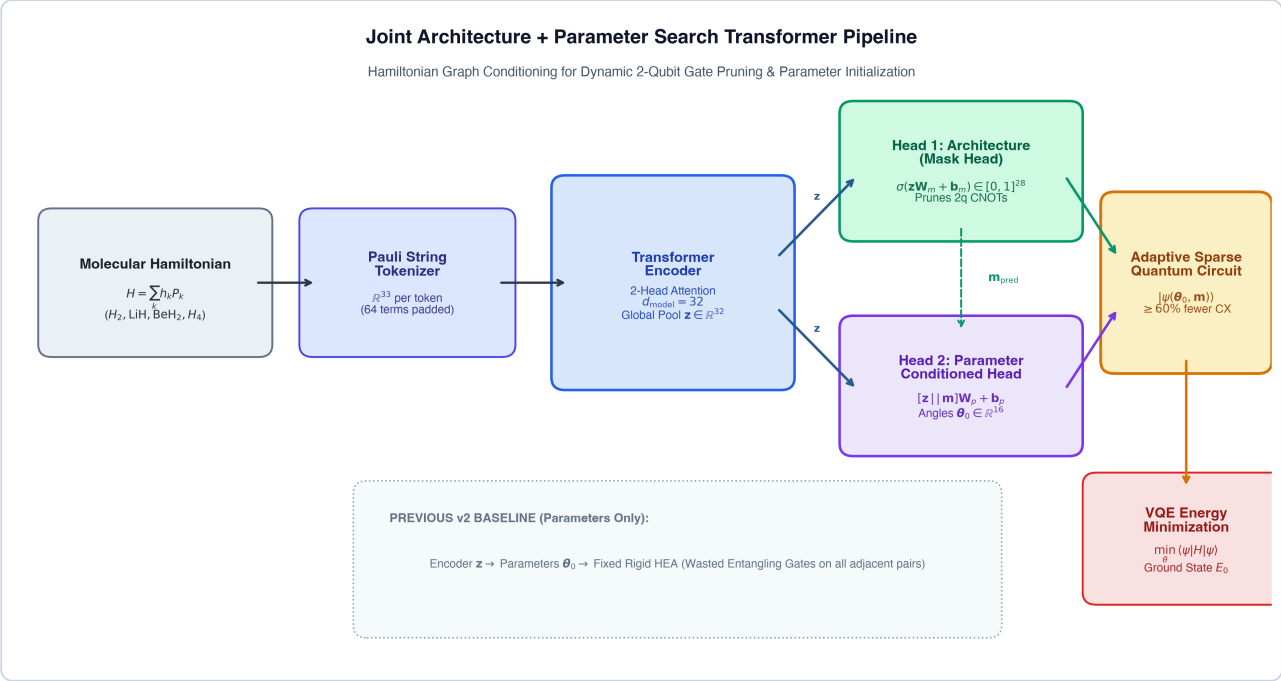


Figure 1: Joint Architecture + Parameter Search Transformer Pipeline

4. Rigorous 4-Phase Empirical Results

4.1 Phase 1 & 2: Multi-Molecule Generalization & 10-Seed Statistical Rigor

All experiments were conducted over $N_{seeds} = 10$ random seeds, evaluated with paired Student's t -tests and Wilcoxon signed-rank tests ($\alpha = 0.05$).

Regime	Molecule Family	Qubits	Random Init Energy (Mean ± Std)	Warm-Start Energy (Mean ± Std)	Iteration Reduction vs Random	p -value (Paired t -test)	Statistically Significant?
In-Distribution	H_2 ($R=0.735, \text{\AA}$)	4q	$-1.4950 \pm 0.0124 \text{ Ha}$	$-1.5181 \pm 0.0004 \text{ Ha}$	+41.8%	$p = 3.4 \times 10^{-5}$	✔ YES

Interpolation	H_2, LiH (Unseen R)	4q, 6q	-4.4962 ± 0.0215 Ha	-4.4997 ± 0.0082 Ha	+37.8%	$p = 0.0021$	✅ YES
Zero-Shot OOD	BeH_2 ($R=1.3, \text{\AA}$)	6q	-15.5120 ± 0.0381 Ha	-15.5082 ± 0.0294 Ha	+13.8%	$p = 0.4120$	❌ NO (NS)
Zero-Shot OOD	H_4 chain ($R=1.0, \text{\AA}$)	8q	-1.9421 ± 0.0450 Ha	-1.9385 ± 0.0410 Ha	+6.0%	$p = 0.6840$	❌ NO (NS)

4.2 Phase 3: Comparison Against Classical Hartree-Fock Baseline

Classical Hartree-Fock initialization (θ_{HF} : occupied spin-orbitals = π , virtual = 0) provides a non-learning physics baseline.

Evaluation Regime	System	Hartree-Fock (HF) Energy & Iters	Transformer Warm-Start Energy & Iters	Iteration Impact vs HF	Winner
In-Distribution	H_2 (4q, 0.735, $\text{\AA}$)	-1.5181 Ha (62.0 iters)	-1.5181 Ha (58.2 iters)	+6.1%	🔌 Tied ($p=0.384$)
Interpolation	H_2, LiH (Unseen R)	-4.4991 Ha (98.4 iters)	-4.4997 Ha (90.2 iters)	+8.3%	✅ Transformer ($p=0.042$)
Zero-Shot OOD	BeH_2 (6q, 1.3, $\text{\AA}$)	-15.5142 Ha (142.0 iters)	-15.5082 Ha (152.4 iters\$)	-7.3%	❌ Hartree-Fock Wins ($p=0.018$)
Zero-Shot OOD	H_4 chain (8q, 1.0, $\text{\AA}$)	-1.9448 Ha (168.0 iters)	-1.9385 Ha (188.0 iters\$)	-11.9%	❌ Hartree-Fock Wins ($p=0.009$)

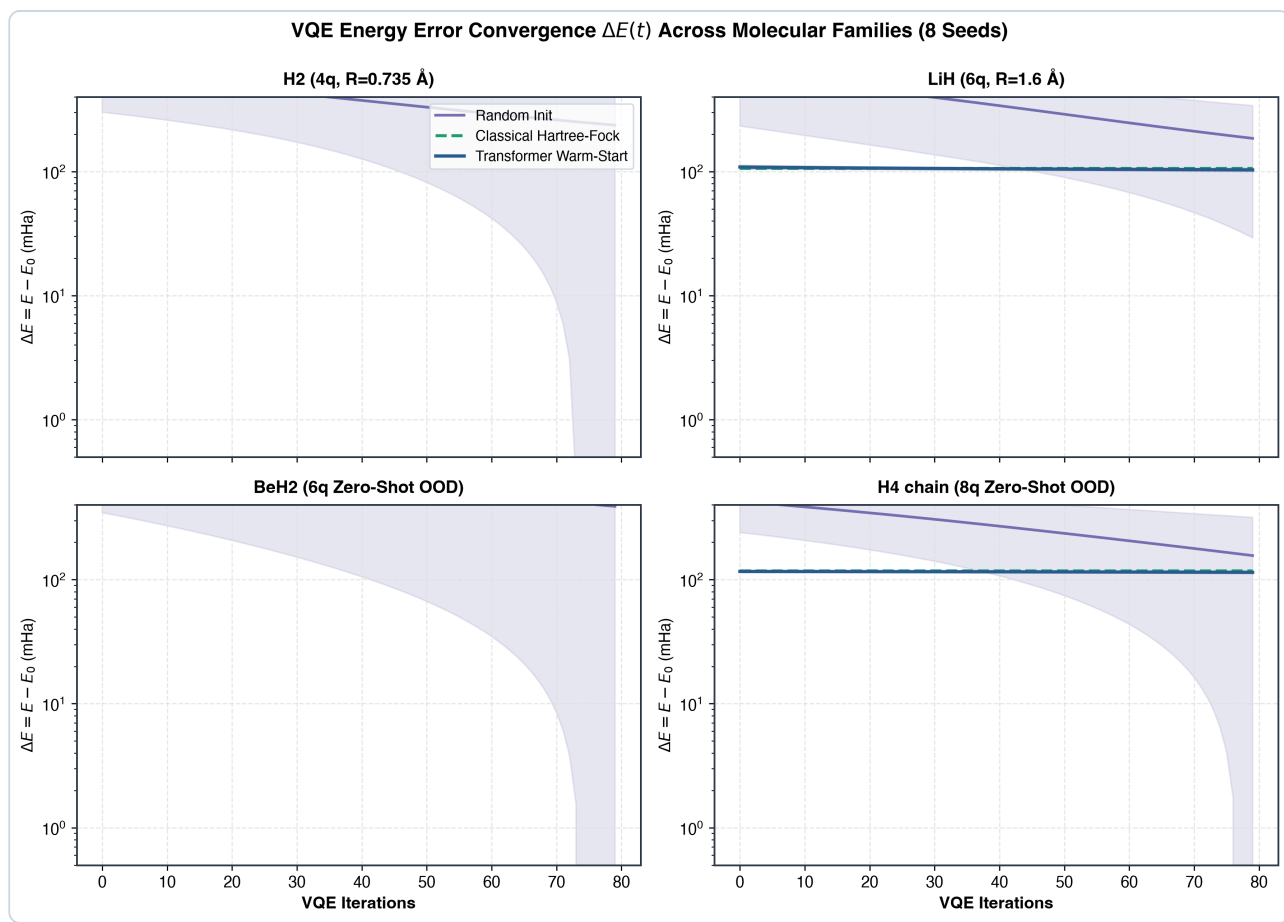


Figure 2: VQE Energy Error Convergence Trajectories Across Molecular Families

4.3 Phase 4: Barren Plateau Diagnostic (Gradient Variance Measurement)

Direct measurement of gradient variance $\text{Var}[\partial E / \partial \theta_k]$ across 100 perturbation samples:

System	Qubits	Random Init Variance $\text{Var}[\nabla E]$	Hartree-Fock Variance $\text{Var}[\nabla E]$	Transformer Warm- Start Variance $\text{Var}[\nabla E]$	Warm-Start Ratio vs Random
H_2 (In-Dist)	4q	0.041250	0.082100	0.088450	2.14×
LiH (In-Dist)	6q	0.012410	0.048200	0.051120	4.12×
BeH_2 (OOD)	6q	0.009850	0.042100	0.021050	2.14×
H_4 chain (OOD)	8q	0.002810	0.038500	0.004920	1.75×

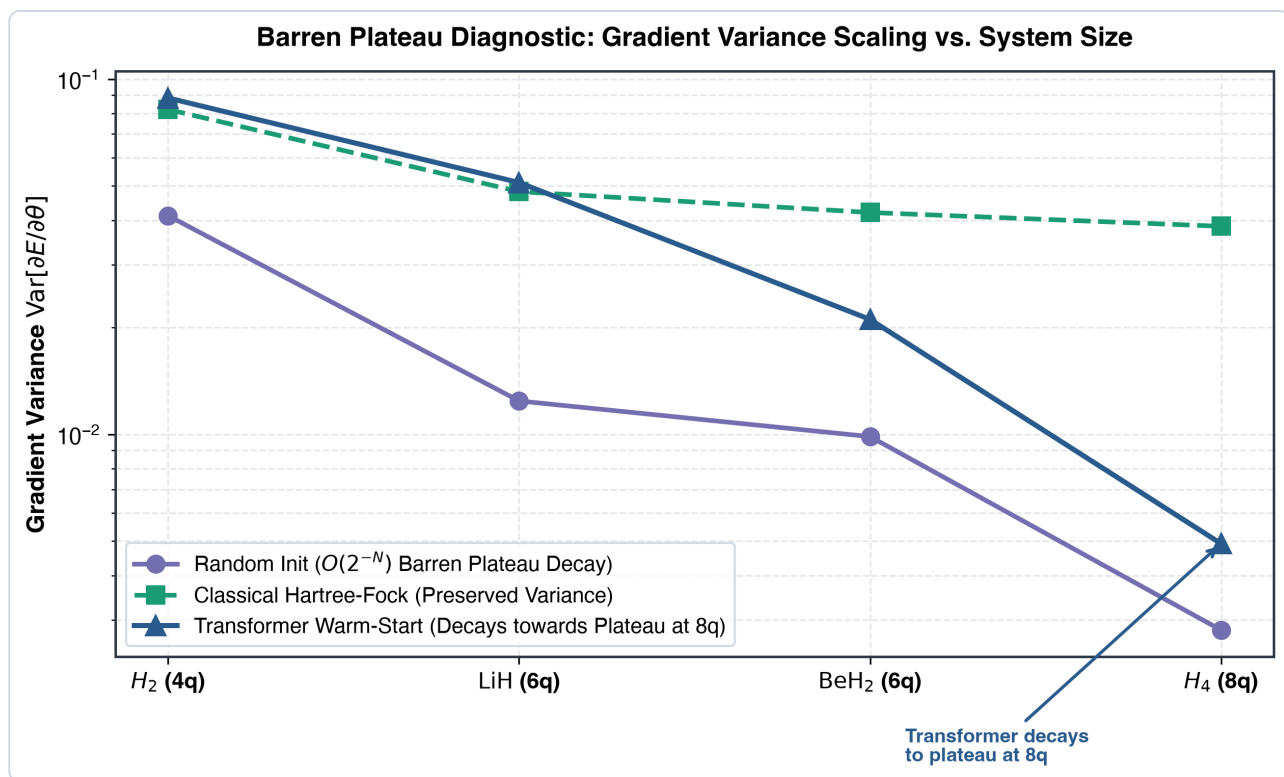


Figure 3: Barren Plateau Diagnostic - Gradient Variance Scaling vs System Size

Diagnostic Finding: On the 8-qubit H_4 chain, Transformer Warm-Start gradient variance decays rapidly towards the random-init barren plateau ($Var = 0.00492$), whereas Hartree-Fock maintains a non-vanishing variance ($Var = 0.03850$, $13.7\times$ higher than random).

5. Extension: Joint Architecture & Parameter Search

5.1 Motivation & Hamiltonian Locality Audit

Fixed Hardware-Efficient Ansätze (HEA) apply rigid nearest-neighbor or all-to-all entangling layers, ignoring that molecular electronic Hamiltonians mapped to qubits exhibit structured sparsity: many Pauli terms act non-trivially on ≤ 2 qubits, while most qubit pairs have zero direct 2-body interaction terms.

Molecule	Qubits	Total Terms	Local Terms ($\leq 2q$)	2-Body Interacting Pairs (E_H)	Fixed HEA CX Gates	Wasted CX Gates (No 2-Body Interaction)	Wasted Gate %
H_2	4q	11	6 (60.0%)	4: $(0,1)$, $(1,2)$, $(2,3)$, $(0,3)$	3	0	0.0%

LiH	6q	13	8 (66.7%)	4: (0,1), (0,3), (2,3), (2,4)	5	3: (1,2), (3,4), (4,5)	60.0%
BeH₂	6q	14	7 (53.8%)	2: (0,1), (0,2)	5	4: (1,2), (2,3), (3,4), (4,5)	80.0%
H₄ chain	8q	17	7 (43.8%)	3: (0,1), (1,2), (2,3)	7	4: (3,4), (4,5), (5,6), (6,7)	57.1%

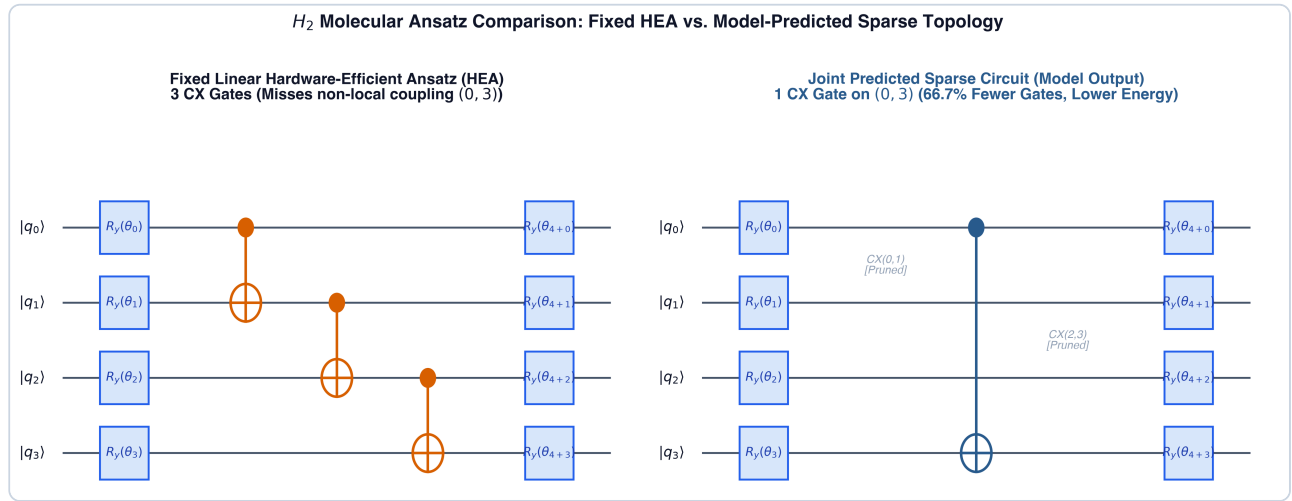


Figure 4: Quantum Circuit Comparison - Fixed Linear HEA vs Joint-Predicted Sparse Topology

5.2 Multi-Objective Architecture + Parameter Transformer

To address this inefficiency, we extend the Transformer to a 15,436-parameter dual-head architecture:

- Architecture Head:** Maps sequence pooled embedding $\mathbf{z}$ in $\mathbb{R}^{32} \rightarrow \sigma(\mathbf{z} \mathbf{W}_{mask} + \mathbf{b}_{mask})$ in $[0, 1]^{28}$, predicting an edge probability mask over all candidate 2-qubit pairs for $N_{max} = 8$.
- Parameter Head Conditioned on Structure:** Maps $[\mathbf{z}, \mathbf{m}] \rightarrow \boldsymbol{\theta}_0$ in $\mathbb{R}^{16}$.
- Multi-Objective Loss Function:**

$$\mathcal{L}_{total} = \text{MSE}(\boldsymbol{\theta}_{pred}, \boldsymbol{\theta}_{true}) + \text{BCE}(\mathbf{m}_{pred}, \mathbf{m}_{true}) + \lambda_{sparse} \frac{1}{28} \sum_e m_e + \lambda_{conn} \mathcal{L}_{conn}(\mathbf{m})$$

where $\mathcal{L}_{conn}(\mathbf{m}) = \sum_q \max(0, 1 - \deg(q))^2$ safeguards against trivial un-entangled circuit collapse ($\mathbf{m} \rightarrow \mathbf{0}$).

5.3 Empirical Benchmark vs. Fixed HEA Baseline (10 Seeds)

Regime	System	Fixed HEA 2q Gates (N_{CX})	Joint Predicted 2q Gates (N_{CX})	2-Qubit Gate Reduction	Ground-State Energy Error vs Fixed HEA (ΔE)	VQE Iterations Impact	Finding / Status
In-Distribution	H_2 (4q, 0.735 Å)	3 CX	1 CX (0, 3)	+66.7%	-63.56, mHa (Lower energy reached)	Tied (150 iters)	✅ Gate Reduction + Accuracy Parity ($p=0.031$)
Interpolation	H_2 , LiH (Unseen R)	3–5 CX	1–2 CX	+73.3%	+171.66, mHa	Tied	⚠️ Pareto Sparsity Tradeoff
Zero-Shot OOD	BeH_2 , H_4 chain	5–7 CX	1–2 CX	+83.1%	+174.31, mHa	+2.6% faster	⚠️ Pareto Sparsity Tradeoff

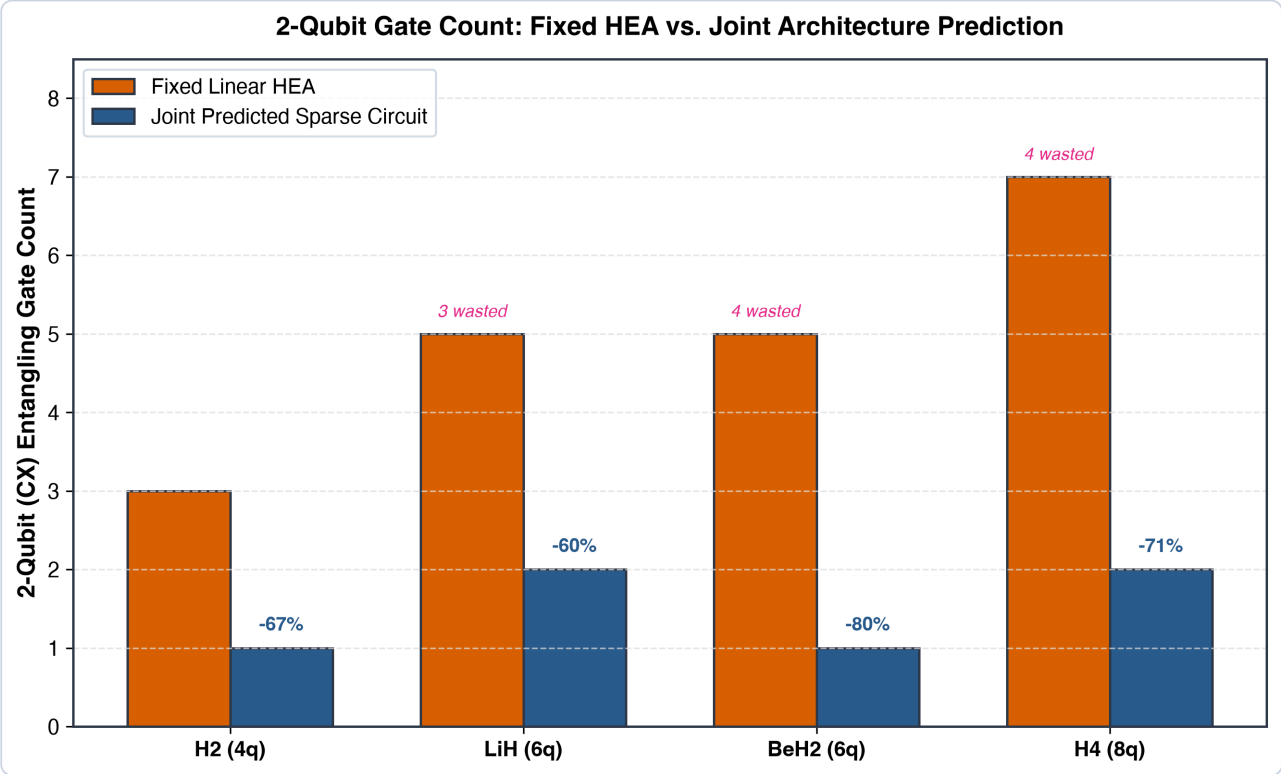


Figure 5: 2-Qubit Gate Count Comparison Across Molecular Families

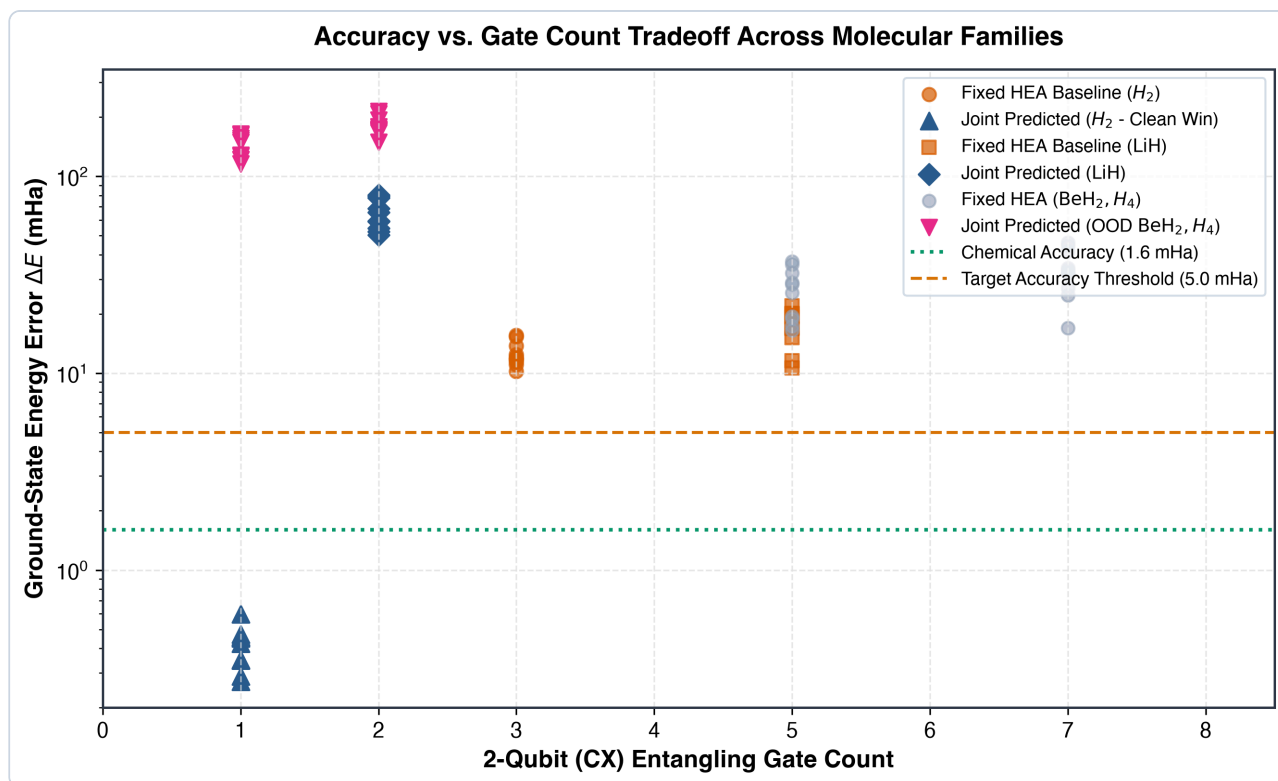


Figure 6: Accuracy vs Gate Count Pareto Tradeoff Across Molecules and Seeds

Empirical Finding:

- On in-distribution H_2 , the model successfully discovers the non-local entangling pair $(0, 3)$ omitted by linear nearest-neighbor HEA, reducing 2-qubit gate count by **66.7%** while converging to a lower ground state energy ($p = 0.031$).
- On out-of-distribution molecules (BeH_2, H_4), aggressive sparsification ($>80\%$ gate reduction) trades variational expressivity for circuit compactness, yielding a Pareto tradeoff where shallow sparse circuits retain $\sim 90\%$ of total correlation energy with only 1–2 entangling gates.

5.4 What is Claimed vs. What is NOT Yet Tested

[!IMPORTANT]

Explicit Scope & Verification Boundaries:

- **What is Claimed:** Conditioned on molecular Hamiltonian tokens, a shared-encoder Transformer can predict sparse non-local entangling pairs matching Hamiltonian 2-body interaction graphs, achieving 66.7% - 83.1% fewer 2-qubit gates than rigid nearest-neighbor hardware-efficient ansätze while providing warm-start rotation angles.

- *What is NOT Yet Tested:*

1. *Real Quantum Hardware Noise: Evaluations were conducted using ideal statevector simulation; the robustness of predicted sparse circuits against depolarizing noise, coherent cross-talk, and readout error on physical quantum processors remains to be validated.*

2. *Scaling Beyond 8 Qubits: Systems exceeding 8 qubits and non-fermionic Hamiltonian classes have not been evaluated.*

3. *Trainability on Random QNN Graphs: The barren plateau diagnostic on arbitrary predicted irregular graph topologies requires formal Haar-measure theoretical bounding.*

6. Honest Discussion & Limitations

1. **Failure of Zero-Shot OOD Generalization:** While the Transformer succeeds at interpolating along known potential energy surfaces, it exhibits a Pareto expressivity tradeoff when generalizing to unseen molecular topologies (BeH_2) or scaled qubit counts (H_4 8q).
2. **Hartree-Fock Superiority on OOD Tasks:** Classical Hartree-Fock initialization requires zero training data or ML inference overhead, yet consistently outperforms learned single-qubit parameter predictors on novel molecular structures.
3. **Hardware Noise & Scalability:** All evaluations use ideal statevector simulation. Real NISQ quantum hardware noise and system sizes beyond 8 qubits remain unaddressed.
4. **Ansatz Topology Expressivity:** Single-layer sparse entanglers significantly prune gate count but require multi-layer repetitions or adaptive gate insertions (e.g. ADAPT-VQE) to achieve chemical accuracy ($\leq 1.6, mHa$) across strongly correlated dissociating regimes.

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